

Nikunj Pradhan

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EDUCATION

Missouri University of Science and Technology
Bachelor of Science, Computer Science

GPA: 4.0/4.0
Aug 2023 - Dec 2026

EXPERIENCE

Data Science Co-Op

Hunter Engineering Company

Jan 2025 – Aug 2025

Bridgeton, MO

- Elevated deep learning model workflow accuracy from 56% to 84% through hyperparameter optimization, active learning, and selective data clustering, improving model reliability for diverse usage.
- Built a CNN-based regression model predicting 8-point image coordinates using RMSE and MAE loss, improving geometric correction accuracy by 23% and streamlining preprocessing.
- Created and maintained custom ETL pipelines using FastAPI and Python to clean, validate, and process metadata and image data from 600+ worksites, ingesting over 2M images into Microsoft SQL Server and AWS S3.
- Automated cross-cloud benchmarking pipelines across AWS and Azure, enabling direct performance comparisons of vendor AI & ML models to drive strategic evaluation and decision-making.
- Reduced model training time by 66% by migrating YOLO detection/classification models to PyTorch Lightning with Distributed Data Parallel for multi-GPU training.
- Delivered full-stack apps (FastAPI, React, SQL Server) production model dashboards, dataset analytical applications, and AI-assisted annotation.

Computer Vision Researcher

Missouri University of Science and Technology

Aug 2023 – Dec 2024

Rolla, MO

- Optimized algorithms, increasing blob analysis accuracy 62% through hyperparameter tuning, and algorithmic refinement using C++ OpenCV and MATLAB.
- Designed a C++/MATLAB application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling with Point Cloud Library.
- Utilized advanced computer vision techniques, including watershed segmentation and 3D model triangulation via ray tracing, for improved object detection and modeling.
- Led weekly research coordination sessions to discuss algorithmic approaches, share updates, conduct code reviews, and plan future experiments.

PROJECTS

Nepali Devanagari OCR Pipeline | *PyTorch, Python, FastAPI*

- Designed and implemented a text detection + CRNN workflow covering 70+ Nepali consonants, vowels, punctuation, and matras, enabling OCR of document, book, and newspaper images.
- Compared CNN backbones and sequence modeling approaches (LSTM vs Bi-LSTM) using both CTC and Attention mechanisms for optimal OCR accuracy.
- Collected and annotated 800+ Nepali text images, applying augmentations to enhance training robustness.
- Implemented FastAPI inference service to deploy models for real-time text recognition pipelines.
- Benchmarked OCR performance against PaddleOCR and EasyOCR on Nepali text datasets.

Movie Recommendation & Watchlist Platform | *FastAPI, React, PostgreSQL, MinIO, Scikit-learn, K3S, Docker*

- Developed a full-stack movie watchlist app with personalized recommendations, allowing users to create, track, rate, and share watchlists.
- Implemented hybrid recommendation system combining content-based filtering and ALS-based collaborative filtering, using per-user caching for fast, accurate suggestions.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, MATLAB, Bash, HTML/CSS

Machine Learning & AI: Pytorch, Scikit-learn, ONNX, CNN/CRNN, Recommendation Systems, Transformers

Databases: Mongoose, PostgreSQL, Microsoft SQL Server, DynamoDB, Firebase, MinIO

Data Science Tools: Pandas, NumPy, SciPy, Matplotlib, Jupyter, OpenCV

Web Development & APIs: FastAPI, React, Node.js, Express.js, WebSockets

Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, Kubernetes, Git, CI/CD, Linux/Ubuntu